

HUDSON C. SMITH

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Track record of taking hard automation problems from ambiguity to working systems — for aerospace, defense, and advanced manufacturing customers — combining deep hands-on technical capability with direct customer ownership and a bias for getting things done. Experienced translating between domain expertise and engineering teams, managing government NRE programs, and building production systems in environments where the playbook doesn't exist yet. Dual M.S. in Mechanical Engineering (Virginia Tech / TU Darmstadt); published researcher in robot safety and ML

EXPERIENCE

GrayMatter Robotics — *Robotics Engineer II — Government Projects*

Los Angeles, CA | Jul 2025 – Present

- Customer-facing technical lead for government NRE programs in robotic weld grinding, ultrasonic inspection, and cold spray repair — source opportunities, write proposals, scope requirements, and own the relationship from first contact through program execution
- Do the hands-on application engineering myself and pull in software resources as needed; serve as the primary bridge between customer mission requirements and internal engineering execution
- Company delegate to the American Foundry Society (AFS); build relationships with defense and manufacturing decision-makers and advance adoption of robotic automation

GrayMatter Robotics — *Robotics Engineer — Systems & Applications*

Los Angeles, CA | Apr 2024 – Jul 2025

- Took commercial aerospace MRO robotic system from R&D into production as the sole on-site engineer — developed scanning parameters, tuned planning configs, wrote SOPs, and trained all customer personnel; owned the customer relationship end-to-end and handed off a live production system as a retained paying account
- Built scan-to-robot abrasive cutting capability from scratch: flew to tooling partner in Texas to learn closed-loop cutting mechanics firsthand, authored the software requirements spec, then drove implementation nights and weekends alongside the engineering team through to customer demonstration
- Led on-site commissioning and live troubleshooting of cement truck refurbishment robotic system through customer handoff
- Conducted technical site assessments at prospective customer facilities to evaluate automation feasibility, surface constraints, and define solution architectures for new programs

ABB Corporate Research — *Robotics Research Engineer (Working Student)*

Ladenburg, Germany | May – Nov 2023

- Applied ML to predict safe stopping distances for industrial robots under ISO/TS 15066 and ISO 10218 — designed and executed experiments on ABB platforms across speeds and payloads, built probabilistic safety models from empirical data
- Published findings and presented at VDI Mechatroniktagung; work constituted the basis of master's thesis: Safe Stopping Distances and Times for Industrial Robotics

GreenbergFarrow — *Mechanical Engineering Designer II*

New York, NY | Dec 2023 – Apr 2024

- Designed MEP systems for Murphy Oil retail sites nationwide in Autodesk Revit; coordinated with civil, structural, and construction stakeholders

GreenbergFarrow — *Mechanical Engineering Intern*

New York, NY | May 2022 – Nov 2023

- Supported MEP system design and documentation across commercial and retail projects; returned full-time after completing dual M.S. degrees

SELECTED PROGRAMS

Robotic Weld Grinding & Ultrasonic Inspection — Defense Manufacturer | NRE

- Technical lead from opportunity sourcing through R&D execution; integrated industrial robot, 3D vision, and force-controlled tooling; managed government stakeholder relationship end-to-end

Aerospace MRO Robotic Automation — Commercial Airline | Deployed

- Sole on-site engineer who took the system from R&D into production: developed scanning parameters, tuned planning configs, wrote SOPs, trained all customer personnel; system remains live under active sustainment contract

Scan-to-Robot Abrasive Cutting Pipeline — Caterpillar | R&D Demonstration

- Technical discovery with tooling partner → authored software requirements → co-developed with engineering team through working customer demonstration; capability served as proof point for subsequent contract award

Safe Stopping Distance Models — ABB Corporate Research | Published

- Probabilistic ML models for robot stopping behavior under ISO 10218/TS 15066; experimentally validated on ABB systems; published at VDI Mechatroniktagung

TECHNICAL SKILLS

Programming: Python (PyTorch, NumPy, OpenCV, Matplotlib), ROS 2, MATLAB, C, C++, VBA

Robotics: Industrial robot programming (ABB, FANUC), robotic grinding & surface finishing, scan-to-path automation, force-controlled manipulation, MRO system design

Vision & Perception: 3D scanning, point cloud processing (PCL), machine vision integration, depth sensor fusion

ML / AI: Supervised learning, regression & classification, experimental design, safety-critical ML deployment

Engineering Tools: ABB RobotStudio, Autodesk Revit, AutoCAD, Inventor, SolidWorks

EDUCATION

Virginia Tech — *M.S. Mechanical Engineering* 2023

Technische Universität Darmstadt — *M.S. Mechanical & Process Engineering* 2023

Hamilton College — *B.A. Physics & German Studies* 2021

HONORS & PUBLICATIONS

- Deutschlandstipendium Scholarship — TU Darmstadt (merit-based, German Federal Government)
- Fulbright ETA Fellowship — Alternate, Germany
- Nottingham Leadership Scholarship — Hamilton College
- Graduate Teaching Assistant — Virginia Tech Physics Department
- Smith, H.C. et al., "Prediction of Stopping Behavior for Robot Safety," VDI Mechatroniktagung, 2023
- Quantifying Squaraine Phase Separation, Aggregate Populations, and Solar Cell Performance with a Correlated Multimethod Approach, Journal of Physical Chemistry C